Deterministic coefficient descent on rational-point bases

2026-11-14. Source `1e3f20752022a341a4612e6a72e8e5690f8c6baa`. All claims unsigned [pcc:unverified].

For a factor base made of actual prime-field points, every successful bounded monic norm pair in the stated model has prime-field coefficients, even when initially sought over an extension field. The proof derives its successful support and branch signs from the norm/divisibility equations. It does not receive them as oracles. The abstract pole-window result has arbitrary arity; the existing four-coefficient chart has a separate exact membership corollary [pcc:unverified].

Central claims: support selection `PCC-20261114-20a4bc`, general rigidity `PCC-20261114-8c21e7`, prime coefficients `PCC-20261114-255fa5`, actual norm divisor `PCC-20261114-0943a7`, actual core `PCC-20261114-5d35b3`. Lean source, prospective contract, exact proof map, editable diagram, SVG, PDF.

Exact general statement

Let L be a field of characteristic a prime p. Let $T \subset L$ be a finite set of distinct public abscissas, all in the prime subfield. Let $f \in L[X]$ have degree at most3 and prime-subfield coefficients. For every $x \in T$, require a public prime-subfield square root of $f(x)$. Thus these are actual rational curve points, rather than arbitrary rational abscissas that may lift only on a twist [pcc:unverified].

Suppose a candidate has

$$A \text{ monic}, \quad \deg A = a, \quad \deg B < a - 1, \quad N = A^2 - fB^2 \mid P_T = \prod_{x \in T} (X - x).$$

Then every coefficient of A and B belongs to the prime subfield. The proof establishes that N is monic of degree $2a$. The source uses natural degrees, whose lower-polynomial guard implies $a \geq 2$. Zero B is not silently excluded. The hypotheses are actual certificate equations and degree guards; no public finder algorithm, favorable closure bound or efficient support-selection assumption appears [pcc:unverified].

This statement is algebraic. Applying its certificates to an elliptic group/scalar relation still uses the existing group bridge and its nonsingularity, distinct-point and label premises. The theorem itself does not manufacture those scalar labels [pcc:unverified].

Why the support and signs need not be supplied

Since P_T splits and has distinct roots, its monic divisor N splits and is separable. The proof takes the roots of N as a finite set S, derives $S \subseteq T$, and proves [pcc:unverified]

$$N = \prod_{x \in S} (X - x), \qquad |S| = 2a.$$

This is a mathematical derivation from divisibility, not an efficient method for finding the unknown divisor N [pcc:unverified].

Let ϕ be a field endomorphism fixing the public abscissas, public square roots and curve coefficients. At a selected node, the norm gives $A(x)^2 = y^2 B(x)^2$. Factoring the difference of squares derives either sign. Because y is fixed, either case gives

$$A(x)\phi(B(x)) - \phi(A(x))B(x) = 0.$$

No nonzero $B(x)$ or successful sign is assumed. Write A^ϕ, B^ϕ for coefficientwise images. The cross polynomial $AB^\phi - A^\phi B$ therefore vanishes on S and has degree below $2a$; it is zero. Mapping the actual norm preserves N. Equal norms and zero cross product give proportional square polynomials. Monic upper normalization forces the scalar to1, hence

$$A^\phi = A, \qquad B^\phi = B.$$

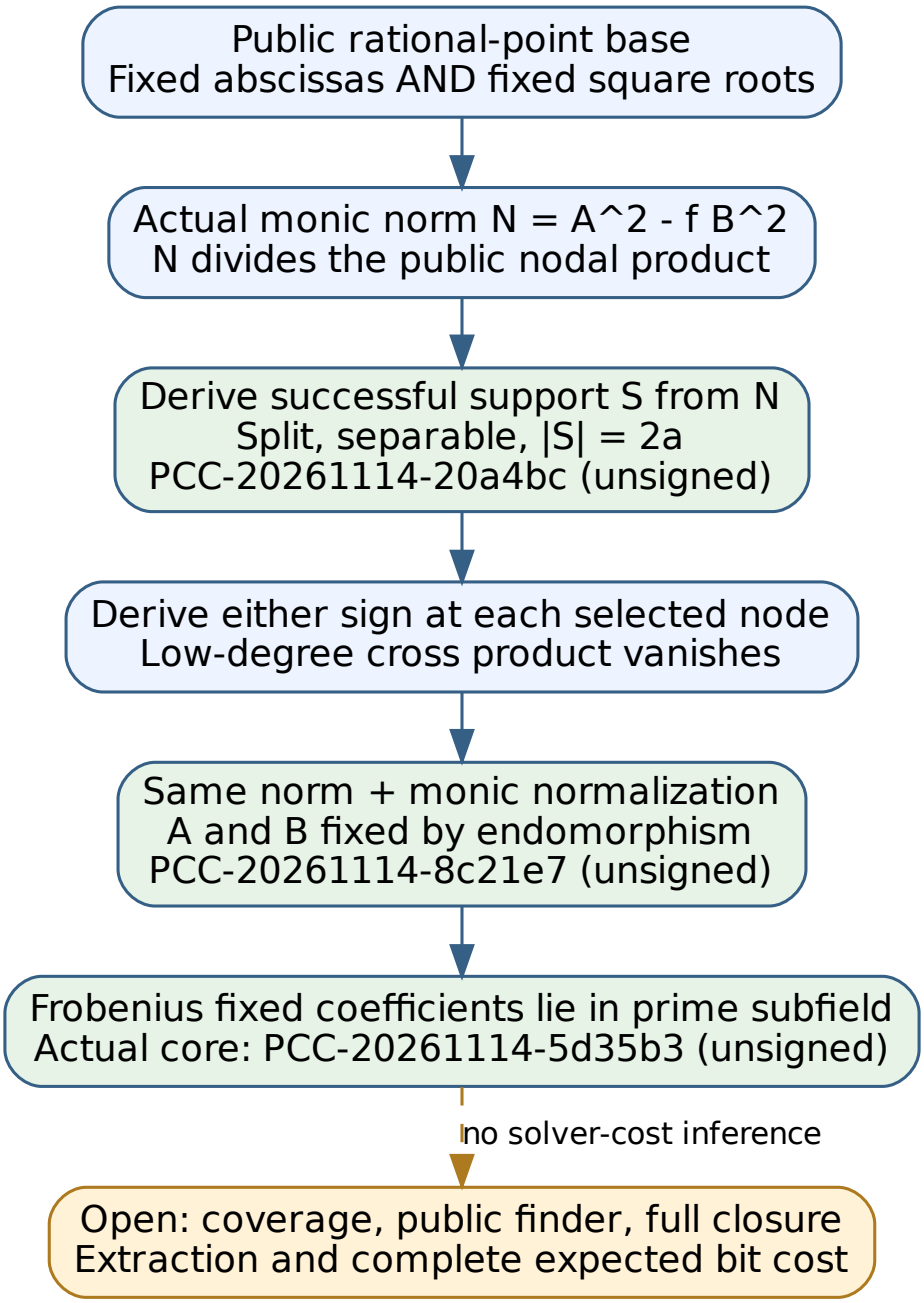


Figure 1: Successful support and signs are consequences of actual norm divisibility on a public rational-point base. Frobenius rigidity then gives prime-field coefficients. No finder or cost bound follows. Evidence unsigned.

The lower polynomial may be zero. This is why neither a division by its value nor a discarded exceptional case is needed [pcc:unverified].

With ϕ equal to Frobenius, every coefficient c satisfies $c^p = c$. Mathlib's exact prime-subfield fixed-point theorem gives $c \in \mathbf{F}_p$. A separate theorem provides literal polynomial lifts to the prime subfield. No agreement-on-sampled-values argument is used [pcc:unverified].

Binding to the existing actual chart

For the current chart, retain the exact target completion

$$D = X^3 + a_2X^2 + a_1X, \quad B = c_0 + c_1X, \quad A = D + zB(r) - D(r), \quad z^2 = r^3 + ar + b.$$

Let the public target abscissa r lie outside the rational base T, and compute the actual monic support

$$U = (A^2 - fB^2)\,\mathrm{div}(X - r).$$

The input equation is the actual remainder $P_T \bmod U = 0$. Its monic division guard yields $U \mid P_T$, hence the full actual norm divides $(X - r)P_T = P_{T\cup\{r\}}$. Target coordinates and every allowed base point are prime-field data. The general result now forces a_1, a_2, c_0, c_1 into the prime subfield. The source proves the upper and lower degree guards and extracts these actual polynomial coefficients [pcc:unverified].

This corollary is the fixed monic degree5 support chart. The abstract rigidity theorem is not restricted to it. Neither statement is a growing-family solver-complexity theorem, nor a claim about arbitrarily redundant free slack variables [pcc:unverified].

The factor-base change and its cost obligation

The base premise is changed openly. It is not a transfer of the earlier Kummer rationality result to Dickson fibers or arbitrary input primes. The following comparison records theorem scope, not measured performance [pcc:unverified].

Item	Earlier Kummer review	Current rational-point base
Allowed abscissas	A Kummer coset; rational lifts may fail	Explicit actual rational points
Rationality mechanism	Good-coset and short mixed-twist relation exclusions	Deterministic norm/Frobenius rigidity
Field conditions for that implication	Includes an admissible 2-power divisor of $p-1$ and other stated premises	Prime characteristic and the explicit rational-point premises
Membership representation	Structured tower	Public nodal product of degree $\lvert T\rvert$; no cheap tower assumed
Finder and complete cost	Unresolved	Unresolved

Known scalar multiples of the input generator are a prospective way to obtain actual rational base points with labels in a fresh attempt. This is not a novel generator claim or free reusable precomputation. Base construction, duplicate/opposite/target exclusions, membership encoding, coverage, failed attempts and solver work must all be charged. No such coverage or cost bound is established by this component [pcc:unverified].

The next central obligation remains a constructive public finder with bounded growing-family normal forms or closure. This descent result removes the mixed-twist field-of-definition issue for this base; it does not turn an algebraic-closure solver into an efficient algorithm or bound its extraction work [pcc:unverified].

Sources, validation and retained failures

Sources checked (KB): unavailable as already recorded, no unchanged retry. The bounded receipt records actual source paths and query scope. Read the independent Kummer rationality review, the actual norm uniqueness sources, and pinned Mathlib split/separable, nodal and prime-subfield declarations. Earlier elliptic-code and non-Semaev reports

remain scoped to their received words and generic-target-access assumptions. These standard ingredients and bounded searches establish no novel algorithm or exhaustive absence [pcc:unverified].

Final module build007 exits0 with no warnings and14 authored statements; only standard foundations are printed. The whole-study gate audits383 statements with0 additional trusted uses, exit0. Import-inclusive fresh replay exits0. Validation receipt PCC-20261114-7b5216 pins source and raw logs in validation-001.json. All15 claims are unsigned/default not_covered, exit2 [pcc:unverified].

All failed drafts001 through006 remain. They record protected-map and identity-wrapper errors, a redundant tactic, an explicit characteristic parameter, finite-set equality instance, constant-power simplification and an omitted existing call argument. No statement was weakened, check suppressed or trusted axiom added. No Rust/native source changed and no prior native tests are counted again [pcc:unverified].

No fixture, unknown-root, F4/msolve, paid/cloud/attack or timing experiment was launched. The earlier native fixture remains1 planned,0 attempted, 0 completed,0 excluded,1 admission-blocked by unchanged checkpoint upload transport. Local universal proof construction is separate. Global performance/curve/attack-route graphs stay unchanged: no measured speedup or new group-transport edge is claimed [pcc:unverified].

Exact proof map

All bindings unsigned [pcc:unverified]. Exact declarations/context hashes and source freeze are retained in claim-pins.json.

Declaration	Claim
`branchCross`	`PCC-20261114-51b429`
`evalMapFixed`	`PCC-20261114-955ed2`
`nodalMapFixed`	`PCC-20261114-e742aa`
`crossFromNorm`	`PCC-20261114-a916b6`
`monicPairUnique`	`PCC-20261114-401e74`
`pairFixedOnNodes`	`PCC-20261114-3d0e01`
`divisorSupport`	`PCC-20261114-20a4bc`
`normWindow`	`PCC-20261114-1f4820`
`pairFixedFromDivisor`	`PCC-20261114-8c21e7`
`frobeniusFixedIff`	`PCC-20261114-68c109`
`pairPrimeCoefficients`	`PCC-20261114-255fa5`
`primePolynomialLift`	`PCC-20261114-bde010`
`actualNormDivides`	`PCC-20261114-0943a7`
`actualCorePrimeCoefficients`	`PCC-20261114-5d35b3`